EDA Analysis

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Data Overview

This exploratory analysis uses `final_data.csv`, generated by `Script/data_script.r`. The dataset brings together four mortality outcomes, conflict and disaster records, and socioeconomic and environmental covariates. Each observation represents a country in a given year, identified by `iso` and `year`.

Across all 17 variables, approximate 86% observations have complete data. Maternal mortality is missing for 426 observations (11.45%), while GDP is missing for 62 (1.67%).

The dataset contains 3,720 observations from 186 countries between 2000 and 2019. Every country has 20 annual observations, forming a balanced panel despite missing values in some variables. All country-year identifiers are complete and unique.

Variables and measurement

The 17 variables comprise two identifiers, one conflict exposure, four mortality outcomes, two disaster indicators, and eight additional covariates.

Table 1: Variables and missing observations in the analysis dataset.

Variable	Role	Description	Missing (n)
iso	Identifier	Three-letter country code	0
year	Identifier	Outcome calendar year	0
armed_conflict	Exposure	Prior-year conflict indicator (0/1)	0
maternal_mortality	Outcome	Maternal mortality ratio	426
infant_mortality	Outcome	Infant mortality rate	20
under5_mortality	Outcome	Under-5 mortality rate	20
neonatal_mortality	Outcome	Neonatal mortality rate	20
drought	Covariate	Drought occurrence indicator (0/1)	0
earthquake	Covariate	Earthquake occurrence indicator (0/1)	0
gdp_1000	Covariate	GDP per capita, scaled in thousands	62
oecd	Covariate	OECD membership indicator (0/1)	0
pop_dens	Covariate	Population density	20
urban	Covariate	Urban population measure	20
age_dep	Covariate	Age dependency ratio	0
male_edu	Covariate	Male educational attainment	20
temp	Covariate	Temperature	20
rainfall_1000	Covariate	Rainfall, scaled in thousands	20

The `armed_conflict` indicator equals 1 when a country records at least 25 deaths across all conflict events in the previous year, and 0 otherwise. For example, conflict in 1999 is linked to mortality outcomes in 2000. Under this definition, 704 observations are classified as exposed to conflict.

Drought and earthquake indicators record whether at least one such event occurred in the outcome year. Each indicator is coded as 1 when an event is recorded and 0 otherwise, regardless of the number of events.

Distributions and summary statistics

Table 2: Summary of continuous variables; N excludes missing observations.

Variable	N	Mean	SD	Median	Min	Max
maternal_mortality	3294	210.62	303.78	66.00	2.00	2480.00
infant_mortality	3700	28.90	26.41	18.90	1.60	138.10
under5_mortality	3700	40.50	42.37	22.20	2.00	224.90
neonatal_mortality	3700	16.18	12.98	12.10	0.80	60.90
gdp_1000	3658	11.49	17.35	4.07	0.11	123.68
pop_dens	3700	30.57	20.75	27.52	0.00	99.86
urban	3700	30.69	17.63	30.25	0.10	93.41
age_dep	3720	61.94	18.85	55.51	16.17	111.48

Table 2: Summary of continuous variables; N excludes missing observations.

Variable	N	Mean	SD	Median	Min	Max
male_edu	3700	8.26	3.02	8.37	1.07	14.44
temp	3700	19.63	7.33	21.96	-2.40	29.68
rainfall_1000	3700	1.20	0.81	1.01	0.02	4.71

The histograms in Figure 1 illustrate the concentration of mortality observations at lower values andthe long tails toward higher mortality. Mortality and GDP have right-skewed distributions, with most observations concentrated at lower values and a smaller number extending into a long upper tail. This pattern is particularly pronounced for maternal mortality, which has a mean of 210.6, a median of 66.0, and a maximum of 2,480.0. Its standard deviation of 303.8 indicates substantial variation across country-years. Because high values pull the mean upward, the median provides a useful description of the centre of the distribution.

The middle 50% of maternal mortality observations lie between 17.0 and 299.8. The 95th percentile is 836.4, meaning that approximately 5% of observed values exceed this level. Together with the much higher maximum, these figures show that the upper tail extends well beyond the values recorded for most country-years. This variation would be difficult to appreciate from the mean alone.

The child mortality outcomes follow a similar pattern. Mean infant, under-5, and neonatal mortality are 28.9, 40.5, and 16.2, respectively, compared with medians of 18.9, 22.2, and 12.1. The middle 50% of observations range from 7.6 to 44.5 for infant mortality, 9.0 to 61.3 for under-5 mortality, and 4.9 to 25.3 for neonatal mortality. Each distribution therefore contains considerable variation, even among observations away from the extremes.

The upper tails also differ across the child mortality outcomes. Their 95th percentiles are 83.3 for infant mortality, 131.6 for under-5 mortality, and 41.2 for neonatal mortality. Under-5 mortality spans a broader numerical range, consistent with its longer age window. However, numerical ranges should be interpreted alongside the definition and scale of each outcome, rather than used alone to judge which outcome is more variable.

GDP is also concentrated toward lower values, with a mean of 11.49 and a median of 4.07 in the supplied thousands scale. Its middle 50% ranges from 1.24 to 13.15, while its 95th percentile is 48.69. This pattern indicates that a relatively small share of country-years has GDP values far above the median. A logarithmic scale could help display this variation in later plots, although the current summary retains the original scale.

The tables use available observations for each variable, so sample sizes differ where values are missing. Each country-year receives equal weight; the summaries are not weighted by population. Conflict exposure is recorded in 704 of 3,720 observations (18.9%), meaning that the exposed group is considerably smaller than the unexposed group.

Table 3: Frequency of binary indicators among nonmissing observations.

Variable	N	Present	Percent
armed_conflict	3720	704	18.9
drought	3720	325	8.7
earthquake	3720	310	8.3
oecd	3720	636	17.1

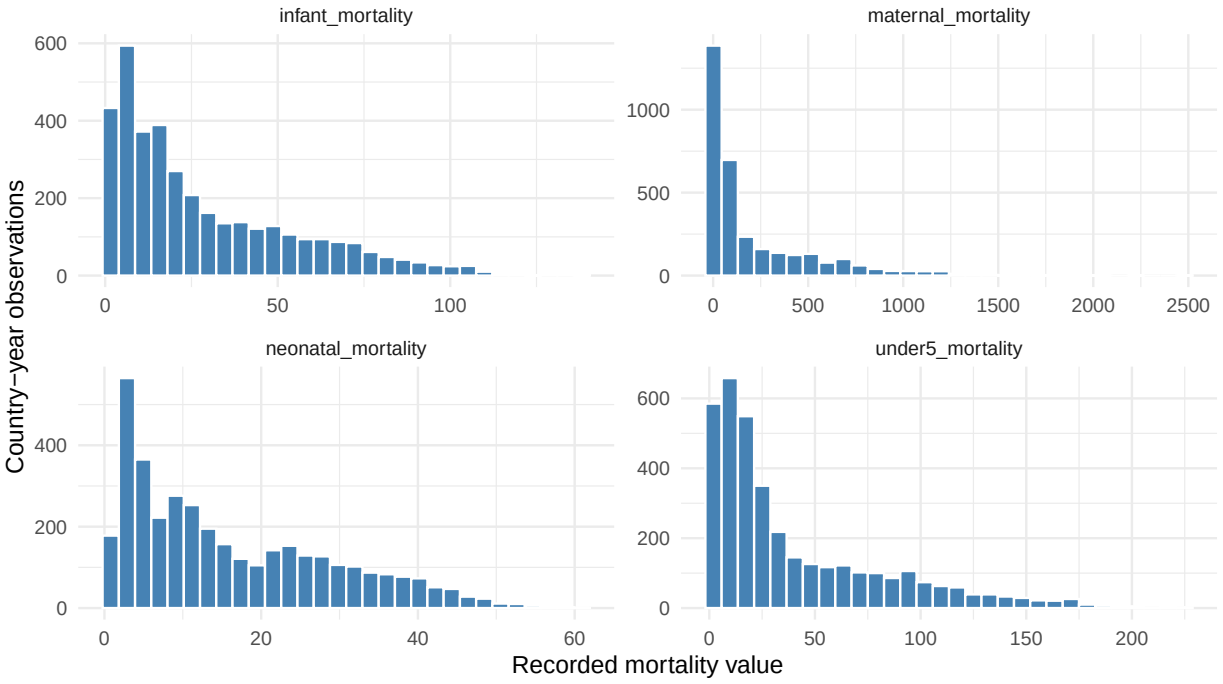

Figure 1: Mortality distributions on their original scales. Panels use separate axes; missing observations are excluded.

Relationships between variables

Table 4: Unadjusted mean mortality and available observations by prior-year conflict exposure.

Outcome	N_Unexposed	N_Exposed	Mean_Unexposed	Mean_Exposed
infant_mortality	2996	704	24.9	46.0
maternal_mortality	2668	626	166.9	396.8
neonatal_mortality	2996	704	14.0	25.2
under5_mortality	2996	704	34.3	66.9

All four mortality outcomes have higher means among observations exposed to conflict in the previous year. Mean maternal mortality is 396.8 in the exposed group, compared with 166.9 in the unexposed group—approximately 2.4 times as high. Mean infant mortality is 46.0 versus 24.9, while under-5 mortality is 66.9 versus 34.3 and neonatal mortality is 25.2 versus 14.0. The consistency of this pattern across outcomes makes the relationship between conflict and mortality worth examining further.

These comparisons use the available observations for each outcome. For example, the maternal mortality means are based on 626 exposed and 2,668 unexposed observations, whereas the child mortality means use 704 exposed and 2,996 unexposed observations. Differences in data availability should therefore be considered when comparing results across outcomes.

The four mortality outcomes are strongly positively related with country-years with higher values for one outcome tend to have higher values for the others. Infant and under-5 mortality have the strongest correlation (0.998). Their overlapping age ranges help explain this close relationship, since deaths before age 1 are also included among deaths before age 5.

The red cells show that higher GDP and greater male educational attainment are associated with lower mortality across all four outcomes. Both variables have a Spearman correlation of -0.83 with maternal mortality, while GDP and male education are positively correlated with each other (0.74). These relationships suggest that socioeconomic conditions should be considered when assessing the association between conflict and mortality. They do not establish an independent effect of either GDP or education.

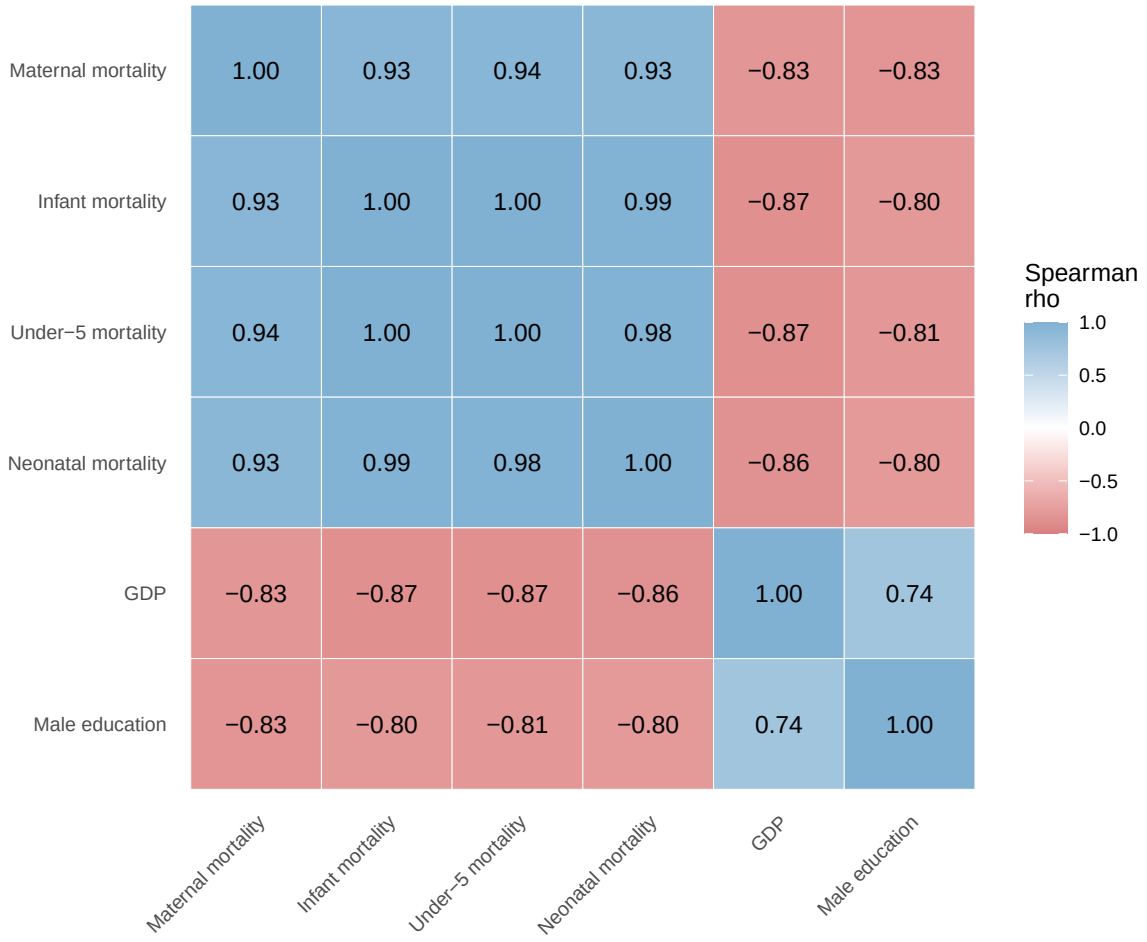

Figure 2: Heatmap Cross Variable.

EDA Findings

The analysis reveals substantial variation in mortality across countries and years. The right-skewed distributions show that a relatively small number of observations have much higher mortality than the rest, making medians and distribution plots useful complements to means.

Mortality is consistently higher among observations exposed to conflict in the previous year. At the same time, the heatmap shows strong relationships between mortality and other socioeconomic conditions. The observed conflict differences may therefore reflect a combination of conflict exposure, economic and educational conditions, and other country characteristics.

The next stage of analysis should account for repeated observations within countries and changes over time. Dealing with missingness should be consider for higher analysis accuracy as an only 3,205 of 3,720 observations are complete across all variables, so restricting every analysis to complete

cases would exclude 515 observations. Meanwhile model-specific sample sizes and the characteristics of excluded observations should be examined before deciding how to handle missing data. These steps would provide a stronger basis for assessing whether the conflict–mortality association persists after adjustment.